

STAT 801A Lab 3: Inference About a Single Mean

Solutions

2026-09-21

```
library(ggplot2)
```

Inference About a Single Mean

In previous work, the parameters for our probability distributions were assumed to be known. Real data does not typically come with its parameters attached, so we must estimate them from the data we do have. We have a sample, and we want to say something about the population it came from. That is inference.

Two questions drive this lab:

- Confidence intervals: what values of μ are consistent with our data?
- Hypothesis tests: is our data consistent with a specific claim about μ ?

Throughout this lab we will assume σ is known.

Notation

Symbol	Meaning	Known?
μ	population mean	No. This is what we want.
σ	population standard deviation	Yes, assumed known
$\bar{y}$	sample mean	Yes
n	sample size	Yes
μ_0	the value of μ a hypothesis claims	Yes, it is given

The sampling distribution of $\bar{y}$

Take a sample, calculate $\bar{y}$. Take another, get a different number. The sample mean is itself a random variable with a distribution.

$$\bar{y} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Its standard deviation is called the standard error.

$$SE = \frac{\sigma}{\sqrt{n}}$$

Confidence Intervals

The sample mean $\bar{y}$ is a point estimate of μ . But a point estimate on its own says nothing about how precise it is, and $\bar{y}$ is almost certainly not exactly μ . A confidence interval reports a range of plausible values instead of a single number.

The formula

$$\bar{y} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

The critical value comes from the standard normal. Calling `qnorm` with no `mean` or `sd` gives $N(0, 1)$.

Note: a 95% interval puts 2.5% in each tail, so use `qnorm(0.975)`, not `qnorm(0.95)`

```
qnorm(0.975)
```

```
[1] 1.959964
```

```
qnorm(0.95) # 90% CI
```

```
[1] 1.644854
```

```
qnorm(0.995) # 99% CI
```

```
[1] 2.575829
```

Computing one in R

A machine fills bags of coffee with a target weight of 340 grams. The population standard deviation is known to be 2 grams. An inspector weighs 20 bags.

```
weights <- c(338.1, 341.2, 339.5, 342.0, 337.8, 340.6, 336.9, 343.1,  
            339.9, 338.4, 341.7, 340.2, 337.5, 342.8, 339.1, 340.8,  
            338.6, 341.4, 336.2, 340.0)
```

```
n <- length(weights)  
ybar <- mean(weights)  
sigma <- 2
```

```
n
```

```
[1] 20
```

```
ybar
```

```
[1] 339.79
```

```
se <- sigma / sqrt(n)  
crit <- qnorm(0.975)  
margin <- crit * se
```

Calculate interval

```
ybar - margin
```

```
[1] 338.9135
```

```
ybar + margin
```

```
[1] 340.6665
```

We are 95% confident the true mean fill weight is between 338.91 and 340.67 grams.

What 95% confidence means

The interval either contains μ or it does not. Once the data is collected there is no probability about it.

The 95% describes the procedure, not this one interval: if we repeated the whole process many times, about 95% of the intervals would contain μ .

Simulating 100 confidence intervals

```
set.seed(2021)

ybars <- replicate(100, mean(rnorm(30, mean = 340, sd = 2)))

margin <- qnorm(0.975) * 2 / sqrt(30)

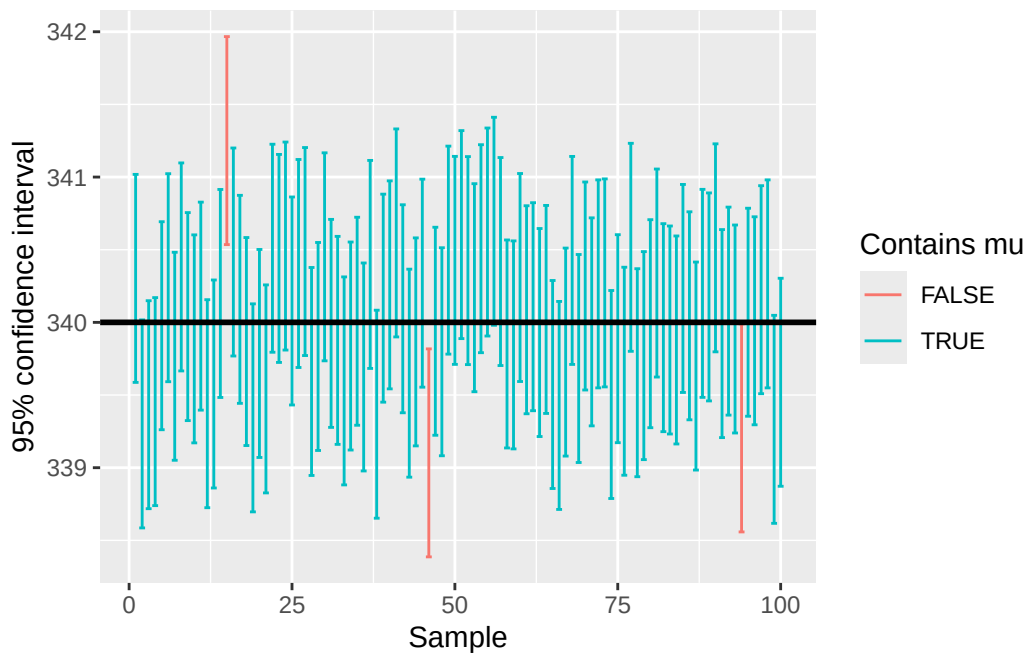
ci <- data.frame(sample_id = 1:100,
                 lower = ybars - margin,
                 upper = ybars + margin)

ci$contains <- ci$lower <= 340 & ci$upper >= 340
mean(ci$contains)
```

```
[1] 0.97
```

96 of the 100 simulated confidence intervals include the real value for μ . This number will be close to 95% every time the simulation is run.

```
ggplot(ci, aes(x = sample_id, ymin = lower, ymax = upper, color = contains)) +
  geom_errorbar() +
  geom_hline(yintercept = 340, linewidth = 1) +
  labs(x = "Sample", y = "95% confidence interval", color = "Contains mu")
```



Note: every interval has the same width, since σ and n never change. Only the center moves. An interval misses when its sample mean lands too far from μ .

Hypothesis Testing

H_0 -> the null hypothesis, the claim we assume is true

H_a -> the alternative hypothesis, what we would like to test

For a single mean the null always has this form.

$$H_0 : \mu = \mu_0$$

The alternative takes one of three forms, depending on the question being asked.

$$H_a : \mu \neq \mu_0 \quad H_a : \mu < \mu_0 \quad H_a : \mu > \mu_0$$

The Null Distribution

The null distribution tells us what to expect when H_0 is true. Everything in a hypothesis test is a comparison against it. The rejection region is its most extreme 5%.

We can build it by simulation -> pretend the null is true, draw many samples from it, and look at the results.

The machine is supposed to average 340 grams, with σ known to be 2. We take samples of 36 bags.

```
mu0 <- 340
sigma <- 2
n <- 36

ybars <- replicate(5000, mean(rnorm(n, mean = mu0, sd = sigma)))

sim_y <- data.frame(ybar = ybars)

head(sim_y)
```

```
      ybar
1 339.6837
2 340.5005
3 339.5770
4 340.1562
5 339.9003
6 340.5501
```

Each value in `sim_y` is the mean of one simulated sample of 36 bags, assuming the null hypothesis is true.

```
se <- sigma / sqrt(n)

lower_cut <- mu0 - qnorm(0.975) * se
upper_cut <- mu0 + qnorm(0.975) * se

lower_cut
```

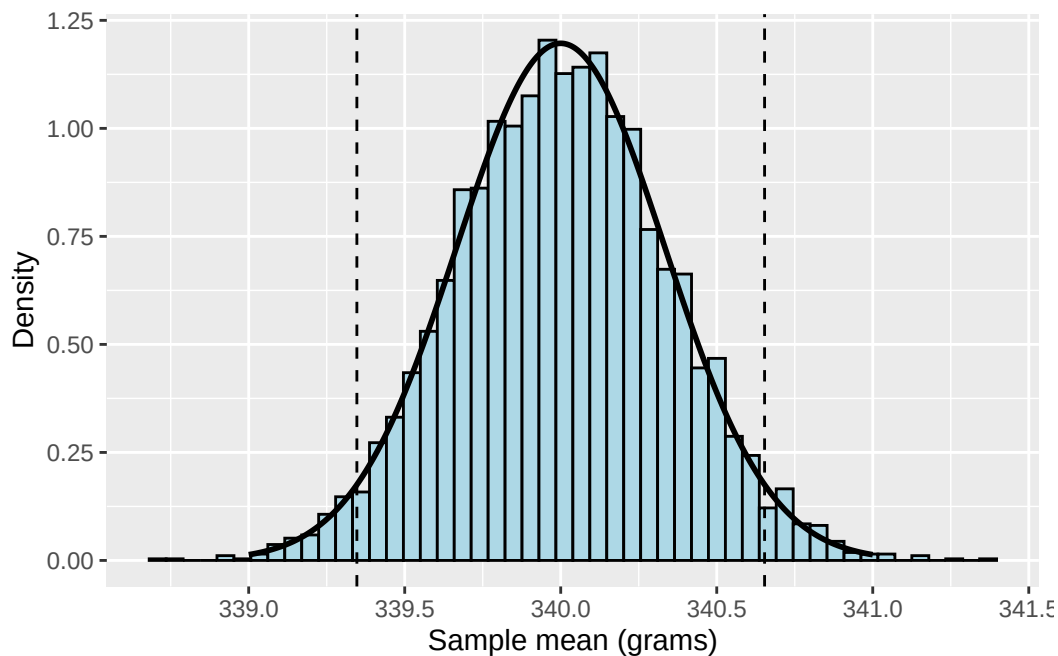
```
[1] 339.3467
```

```
upper_cut
```

```
[1] 340.6533
```

```
theory_y <- data.frame(x = seq(339, 341, length.out = 500))
theory_y$density <- dnorm(theory_y$x, mean = mu0, sd = se)

ggplot(sim_y, aes(x = ybar)) +
  geom_histogram(aes(y = after_stat(density)), bins = 50,
    fill = "lightblue", color = "black") +
  geom_line(data = theory_y, aes(x = x, y = density),
    linewidth = 1, inherit.aes = FALSE) +
  geom_vline(xintercept = c(lower_cut, upper_cut), linetype = "dashed") +
  labs(x = "Sample mean (grams)", y = "Density")
```



The simulated sample means follow the theoretical curve $N(\mu_0, \sigma^2/n)$.

The test statistic

$$z = \frac{\bar{y} - \mu_0}{\sigma/\sqrt{n}}$$

The numerator is how far our sample mean sits from the null. The denominator is the standard error.

So z answers: how many standard errors is our sample mean from the null value?

When H_0 is true, this statistic follows the standard normal.

$$z \sim N(0, 1)$$

Rejection regions

The machine is supposed to average 340 grams, with σ known to be 2. A sample of 36 bags averages 339.1. Test at $\alpha = 0.05$ whether the machine is off target.

$$H_0 : \mu = 340 \quad H_a : \mu \neq 340$$

```
mu0 <- 340
sigma <- 2
n <- 36
ybar <- 339.1

z <- (ybar - mu0) / (sigma / sqrt(n))
z
```

```
[1] -2.7
```

The rejection region is the set of test statistics extreme enough to reject. A two-tailed test at $\alpha = 0.05$ splits the 5% between both tails.

```
qnorm(0.025)
```

```
[1] -1.959964
```

```
qnorm(0.975)
```

```
[1] 1.959964
```

We reject if $z < -1.96$ or $z > 1.96$. Our z is -2.7, so we reject H_0 and conclude the machine is off target.

The null distribution on the z scale

Standardizing every simulated sample mean turns the picture above into the null distribution of z .

```
SE <- sigma / sqrt(n)

z_stats <- replicate(1000, (mean(rnorm(n, mean = mu0, sd = sigma)) - mu0) / SE)

sim <- data.frame(z = z_stats)

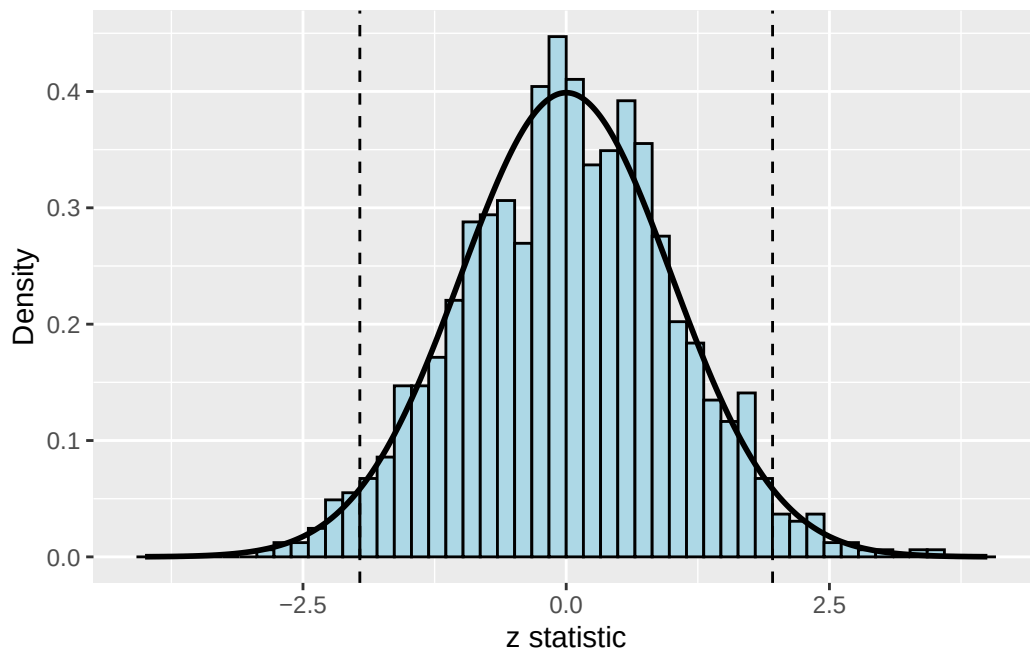
head(sim)
```

```
      z
1 -1.1415858
2  0.1159811
3 -0.3639016
4 -0.5467305
5  1.8286183
6 -1.0067106
```

Same simulation, same samples, different scale. The center is now 0 and the cutoffs are ± 1.96 no matter what the data measures. Different problems will have different data scales, standardizing puts them all on the same one.

```
theory <- data.frame(x = seq(-4, 4, length.out = 500))
theory$density <- dnorm(theory$x)

ggplot(sim, aes(x = z)) +
  geom_histogram(aes(y = after_stat(density)), bins = 50,
                 fill = "lightblue", color = "black") +
  geom_line(data = theory, aes(x = x, y = density),
            linewidth = 1, inherit.aes = FALSE) +
  geom_vline(xintercept = c(-1.96, 1.96), linetype = "dashed") +
  labs(x = "z statistic", y = "Density")
```



```
mean(abs(z_stats) > 1.96)
```

```
[1] 0.051
```

About 5% of simulated z statistics under the null hypothesis fall inside the rejection region. That is what α means: the proportion of times we reject a null that was actually true. This is also known as a type 1 error.

p-values

The rejection region gives a yes or no. A p-value reports how extreme the data was: the probability, assuming H_0 is true, of a test statistic at least as extreme as ours.

Two-tailed, so we take both tails.

```
2 * pnorm(abs(z), lower.tail = FALSE)
```

```
[1] 0.006933948
```

Since $0.0069 < 0.05$, we reject the null

One-tailed tests

If only one direction matters, the alternative points one way.

$$H_0 : \mu = 340 \quad H_a : \mu < 340$$

All 5% goes into the lower tail.

```
qnorm(0.05)
```

```
[1] -1.644854
```

```
pnorm(z)
```

```
[1] 0.003466974
```

Note: the one-tailed critical value is closer to zero, which makes rejecting easier

Alternative	Critical value at $\alpha = 0.05$	p-value in R
$\mu \neq \mu_0$	$\pm \text{qnorm}(0.975)$	$2 * \text{pnorm}(\text{abs}(z), \text{lower.tail} = \text{FALSE})$
$\mu < \mu_0$	$\text{qnorm}(0.05)$	$\text{pnorm}(z)$
$\mu > \mu_0$	$\text{qnorm}(0.95)$	$\text{pnorm}(z, \text{lower.tail} = \text{FALSE})$

Writing a z-score function

R has no built-in z test, so this is a good place to write our own function.

```
z_c <- function(y, mu = 0, sigma = 1) {  
  # Remove missing values  
  count_missing <- sum(is.na(y))  
  message(paste('Number of missing observations that are excluded:', count_missing))  
  
  y <- y[!is.na(y)]  
  
  # Compute from data  
  ybar <- mean(y)
```

```

n <- length(y)

# Compute test statistic
test_statistic <- (ybar - mu) / (sigma / sqrt(n))

# Return value
return(test_statistic)
}

```

Run it on the coffee weights against a null value of 340

```

weights <- c(338.1, 341.2, 339.5, 342.0, 337.8, 340.6, 336.9, 343.1,
             339.9, 338.4, 341.7, 340.2, 337.5, 342.8, 339.1, 340.8,
             338.6, 341.4, 336.2, 340.0)

z_c(weights, mu = 340, sigma = 2)

```

Number of missing observations that are excluded: 0

```
[1] -0.4695743
```

The sample mean of 339.79 sits less than half a standard error below the target

```
z_coffee <- z_c(weights, mu = 340, sigma = 2)
```

Number of missing observations that are excluded: 0

```
2 * pnorm(abs(z_coffee), lower.tail = FALSE)
```

```
[1] 0.6386592
```

Large p-value, so we do not reject. This matches the confidence interval from earlier, which contained 340. A 95% interval and a two-tailed test at $\alpha=0.05$ always agree

Practice Problems

Answer each question in the code chunk provided. Where a problem asks for hypotheses, write them in LaTeX.

Problem 1:

A sample of 25 batteries has a mean lifetime of 487 hours. The population standard deviation is known to be 45 hours.

(a) Compute the standard error.

```
se <- 45 / sqrt(25)
se
```

```
[1] 9
```

(b) Find the critical value for a 95% confidence interval.

```
crit <- qnorm(0.975)
crit
```

```
[1] 1.959964
```

(c) Construct the 95% confidence interval.

```
margin <- crit * se
487 - margin
```

```
[1] 469.3603
```

```
487 + margin
```

```
[1] 504.6397
```

(d) Interpret the interval in a sentence.

We are 95% confident the true mean battery lifetime is between 469.36 and 504.64 hours.

(e) Build a 99% interval. Is it wider or narrower, and why?

```
margin_99 <- qnorm(0.995) * se
487 - margin_99
```

```
[1] 463.8175
```

```
487 + margin_99
```

```
[1] 510.1825
```

The 99% interval runs from 463.82 to 510.18 hours, so it is wider. More confidence requires covering more values, which means a larger critical value (2.576 instead of 1.96).

Problem 2:

A machine fills bags with a target of 340 grams, σ known to be 4. A sample of 36 bags averages 338.5. Test at $\alpha = 0.05$ whether the mean has shifted.

(a) State the hypotheses.

$$H_0 : \mu = 340 \quad H_a : \mu \neq 340$$

(b) Compute the test statistic.

```
z <- (338.5 - 340) / (4 / sqrt(36))  
z
```

```
[1] -2.25
```

(c) Find the rejection region.

```
qnorm(0.025)
```

```
[1] -1.959964
```

```
qnorm(0.975)
```

```
[1] 1.959964
```

We reject if $z < -1.96$ or $z > 1.96$.

(d) Is the test statistic in the rejection region? State your conclusion.

The test statistic is -2.25, which is below -1.96, so it falls in the rejection region. We reject H_0 and conclude the mean fill weight has shifted away from 340 grams.

(e) Compute the p-value and confirm it gives the same conclusion.

```
2 * pnorm(abs(z), lower.tail = FALSE)
```

```
[1] 0.02444895
```

The p-value is 0.024, which is below 0.05, so we reject H_0 . This matches part (d).

Problem 3:

A trainer claims a new program raises mean vertical jump above 24 inches. Fifteen athletes are measured, with σ known to be 1.2 inches.

```
jumps <- c(25.1, 23.8, 26.2, 24.9, 25.5, 22.9, 26.8, 24.2,  
           25.9, 23.6, 27.1, 25.3, 24.7, 26.4, 25.0)
```

(a) State the hypotheses. This is a one-tailed test.

$$H_0 : \mu = 24 \quad H_a : \mu > 24$$

(b) Compute the sample mean and the standard error.

```
ybar_j <- mean(jumps)  
se_j <- 1.2 / sqrt(length(jumps))  
  
ybar_j
```

```
[1] 25.16
```

```
se_j
```

```
[1] 0.3098387
```

(c) Compute the test statistic.

```
z_j <- (ybar_j - 24) / se_j  
z_j
```

```
[1] 3.743884
```

```
z_c(jumps, mu = 24, sigma = 1.2)
```

Number of missing observations that are excluded: 0

```
[1] 3.743884
```

(d) Find the critical value at $\alpha = 0.05$ and state your conclusion.

```
qnorm(0.95)
```

```
[1] 1.644854
```

The test statistic of 3.74 is greater than the critical value of 1.645, so we reject H_0 . There is evidence the mean vertical jump is above 24 inches.

(e) Compute the one-tailed p-value.

```
pnorm(z_j, lower.tail = FALSE)
```

```
[1] 9.05987e-05
```

The p-value is about 0.00009, far below 0.05, which agrees with part (d).

(f) Build a 95% confidence interval for the mean vertical jump.

```
ybar_j - qnorm(0.975) * se_j
```

```
[1] 24.55273
```

```
ybar_j + qnorm(0.975) * se_j
```

```
[1] 25.76727
```

We are 95% confident the true mean vertical jump is between 24.55 and 25.77 inches.

Final Step

Render your document. Click the Render button at the top of the editor, or press Ctrl + Shift + K.

If the render fails, read the error message. It names the chunk that caused the problem. The most common causes are a missing closing parenthesis, or an object used before it was created.